

Continuous Wave Fiber Laser

User Manual

Shenzhen JPT Opto-electronics Co., Ltd

Safety Summary

Thank you for choosing continuous wave (CW) fiber laser from Shenzhen JPT Opto-electronics Co., Ltd.

Please read this manual carefully before attempting to install or operate the laser. This use manual details all the health and safety issues regarding the safe installation and operation. For safety and the best laser performance, please be aware of the laser hazard information and the operational instructions.

- This fiber laser module carries a Class 4 Laser rating. Before connecting it to alternating current (AC) power supply, please make sure that the 220 V AC power supply is connected properly. Wrong connection will spoil the fiber laser.
- Please make sure that a current-controlled AC power supply with proper grounding is used. This is essential to avoid possible personal injury.
- This fiber laser emits invisible laser radiation at 1080 nm with a rated average output power of more than 1000W. Contact with direct or scattered laser radiation can cause permanent damage to the eyes, burn human tissue and start fires.

Users are not allowed to open the casing of the laser. All maintenance and service can only be done at JPT factory. Without the permission from JPT, opening the laser casing will invalidate the warranty.

- Avoid direct eye exposure. Always wear the laser safety goggles when operate the laser.

Table 1 Safety and Compliance Labelling

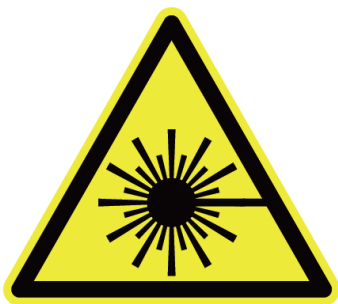
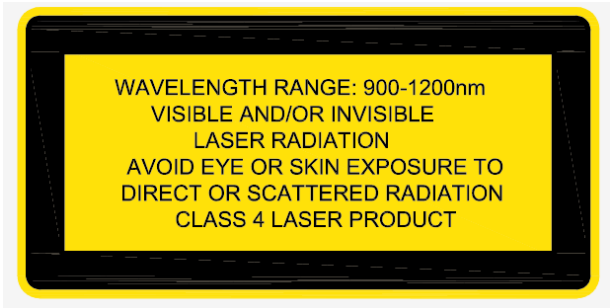
Symbols	Definition
	Laser warning triangle. This symbol is to alert the user to the danger to exposure of hazardous invisible laser radiation.
	Maximum output power of laser



Figure 1 Laser safety goggles

1. Product Tour

Compared to traditional laser systems, fiber lasers possess many advantages, such as high electrical to optical conversion efficiency, compactness, maintenance free, and flexible output enabled by fiber guidance, etc. Fiber lasers are the ideal light source for laser machining, welding and other industrial applications.

High power CW fiber lasers from JPT are the optimal combination of optical, mechanical, electrical and software components. Through the controlling ports and the controlling software, the operating status of the laser can be monitored in real time, alarm messages can be received in time, and data can be collected. The laser uses water-cooling and the power consumption is low. It is also compact and convenient to use.

Below is the picture of a real JPT CW fiber laser product:



Figure 2 Picture of a real fiber laser product

2. Delivered Product

Table 2 Packing list

S/N	Part	Specification	Quantity
1	1000W fiber laser	1000W	1
2	Power cable	5m	1
3	Control cable	5m	1
4	Thumb drive (with controlling software)		1
5	Dust-free tissue, cotton swab		1
6	USB2.0-to-RS232 cable		1
7	User manual		1
8	Keys		2

After unpacking, verify that all components are on hand using the packing list. Please keep all the packaging materials. If long-distance transport is needed again, please reuse the packaging materials in order not to damage the product.

3. General Specifications & Installation Instructions

1) Specifications

A. Optical Characteristics

Table 3

S/N	Characteristics	Testing Condition	Symbol	Min	Typ	Max	Unit
1	Mode of operation			CW/Modulated			
2	Polarization			Random			
3	Nominal output power		P _{nom}	990		1020	W
4	Central emission wavelength	Output power: 1000 W	λ		1080		nm
5	Bandwidth	Output power: 1000 W	$\Delta\lambda$		4	6	nm
6	Short-term power stability	Output power: 1000 W 2-hour continuous operating			1.0	2.0	rms %
7	Long-term power stability	Output power: 1000 W 24-hour continuous operating			±1	±2	%
8	Switch on/off time	Output power: 1000 W				20	μs
9	Modulation frequency					20	kHz
10	Red light power			0.1			mW

B. Optical Output Beam

Table 4

S/N	Characteristics	Symbol	Min	Typ	Max	Unit
1	Output connector	Standard QBH				
2	Beam Parameter Product	BPP	SM 0.45, MM 1.1			mm*mrad
3	Numerical Aperture	NA		0.12		
4	Optical cable length	L		15		m
5	Optical cable bending radius	R	200			mm

C. Mechanical & Environmental Specifications

Table 5

S/N	Characteristics	Min	Typ	Max	Unit
1	Operating temperature	10		40	°C
2	Humidity	10		90	%
3	Storage temperature	- 20		+ 50	°C
4	Dimensions (width × depth × height)	430× 760× 195.5			mm
5	Weight			60	kg
6	Cold start temperature	20			°C

2) Mechanical Details of the Laser Module

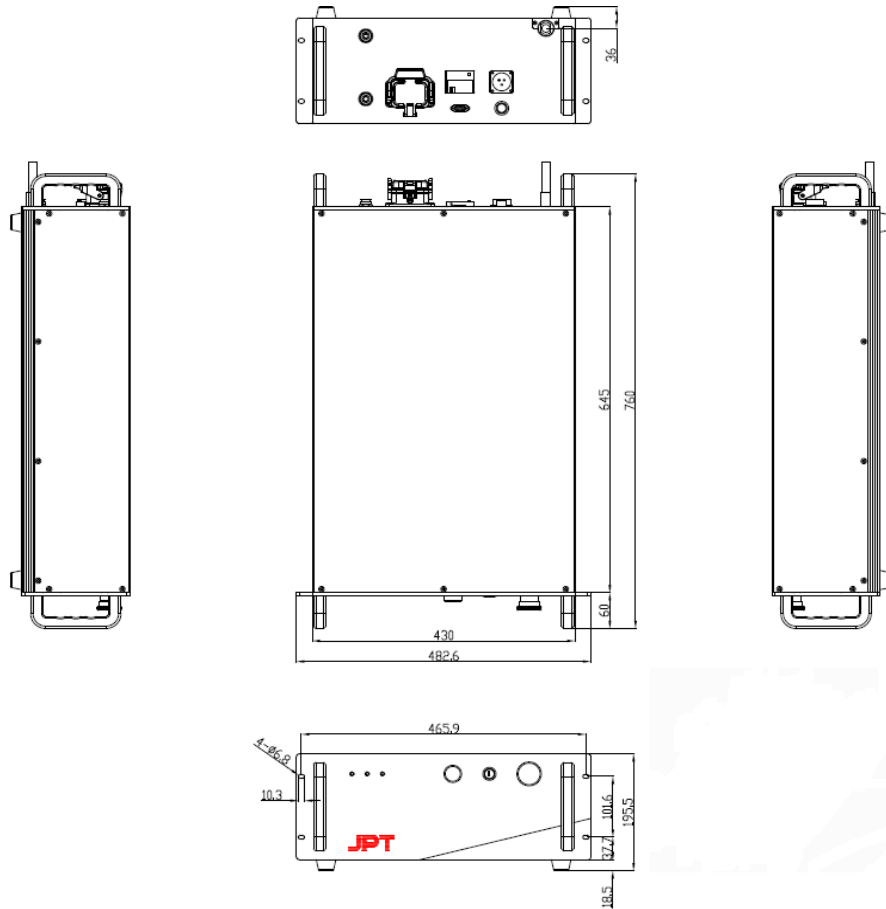


Figure 3 Dimensions of the laser

3) Mechanical Details of fiber connector

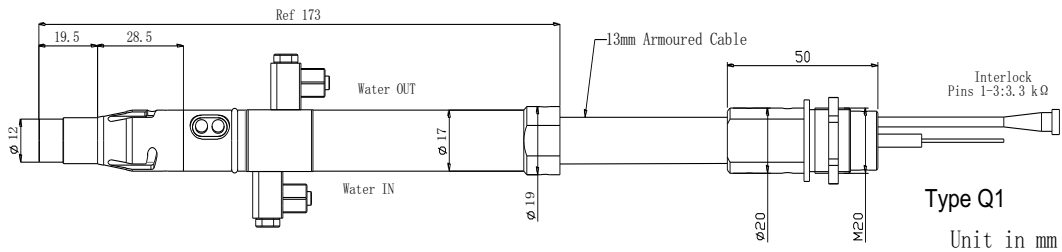


Figure 4 Dimensions of fiber connector

※ the real fiber connector may vary a little from the schematics

4) Mounting and Alignment Procedure

- ※ Make sure that the laser and the fiber connector are placed on a steady horizontal operation platform. If the laser is placed on a metallic platform, then the laser must be in good electrical contact with the platform and the platform should also be grounded properly.
- ※ Make sure that during the installation and the processes afterwards, no component of the laser will be dropped or shaken severely. Beware of falling objects which make damage the laser.
- ※ The metal shield optical cable at the output of the laser should never be bent to a circle with a radius less than 20 cm, no matter in the storage, transport or operating state.
- ※ Before the installation, please make sure that all the power switches and the earth leakage protection are off.

A. Mounting Instructions for the fiber connector

Please firstly put the fiber connector into the external devices if needed (such as cutting head, welding head, etc.). Please make sure that during the whole process is operating environment is clean and the output end facet should not be contaminated. If necessary, special dust-free cotton swab with >99.9% alcohol can be used to clean the output end facet. Any damage to the laser caused by the actions not listed above is not covered by the warranty.

B. Water Cooling System Connection

This laser system uses water cooling and the diameter of the water pipes are ID8/OD12mm. Please note that the inlet and outlet water pipes should be connected as indicated in Figure 5. In order to avoid the abnormal laser operation due to water leakage, the water pipe should not be bent when connecting. More details about the cooling requirement can be found in Table 6.

※ **Important: Before turn on the laser, please start the chiller firstly.**



Figure 5 Inlet and outlet ports for water cooling

Table 6 Water cooling requirement for the laser

S/N	Characteristics	Min	Typ	Max	Unit
1	Cooling method	Water cooling (purified water)			
2	Temperature setting	24	25	26	°C
3	Water pressure			7	bar
4	Required water flow	10			l/min
5	Cooling capacity	5.2			kW

Note: When the laser is not in use for a long time, the water cooling channel in the laser should be emptied and the inlet and outlet ports should be closed using the plugs provided. If the weather is cold, please empty the cooling channel immediately after using the laser, in case it cracks or get blocked.

C. Power Supply Connection

- 1) Before connecting it to the AC power supply for the first time, please make sure that all the power switches, on both the front and back panels, are at the OFF state.
- 2) This laser uses 3PIN aviation plug input port with definition listed in Table 7 and standard listed in Table 8. Please make sure that the electric supply fulfills the requirements before connecting the laser to it. Wrong connection may lead to errors or even damage the laser. The power connection is as shown in Figure 6.



Figure 6 AC power connection

Table 7 AC input connection

S/N	Cables	Electric supply	Color
1	Live wire	L	Brown
2	Neutral wire	N	Blue
3	Earth wire	Protective earth PE	Green/Yellow

Table 8 AC power supply

S/N	Parameters	Values	Units
1	Electric power	220±10%	VAC
2	Current frequency	50/60	Hz

D. I/O Connector and Control Cable

This laser uses 24PIN heavy duty connector with definition listed in Table 9.

The interface connection is as shown in Figure 7.



Figure 7 Interface connection

E. Chiller Connector



Table 9 Chiller connector pin out

PIN	Function	Type	Signal level	Description
1	Chiller signal +	Closed contact input		
7	Chiller signal -			
3				
4				
5				
6				
2				
8				

Note: The female connector will be plugged into the male chiller connector, with pin1 and pin7 been short-circuited before shipping. If necessary, please connect the chiller's closed contact signal to pin1 and pin7.

4. Control and Operation

1) I/O Connection :

Table 10 I/O connection definitions

PIN	Function	Type	Signal level	Description
1				
2				
3	Interlock+	Closed contact input		Reserved
4	Interlock+			
5				
6				
7				
8				
9				
10	Remote start+	Instant closed contact input	24VDC	
11	Remote start-			
12				
13	0-10V+	Analog input	1-10VDC	Power control input: 1-10V=10-100%power
14	0-10-/Analog input			GND for pin 12&13
15	Modulation+	24VDC		PWM modulation
16	Modulation-			
17				
18	Enable+	24VDC		
19	Alarm output	24VDC		Faulty: high signal level
20	Enable -/Alarm output			GND for pin 18 &19
21				
22				
23				
24				
PE	GND			Earth

2) Software Control

A. Function of the Software

Laser control, operation instructions, output mode selection, laser status monitoring, and alarm message, etc.

B. Driver Installation

- 1) Connect the delivered thumb drive to a computer. The thumb drive contains two folders, as shown in Figure 8.

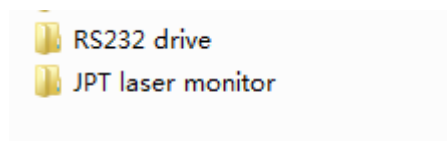
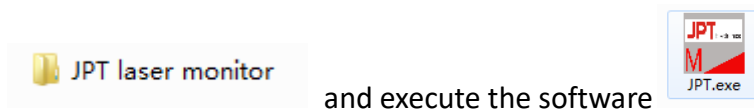


Figure 8

- 2) Firstly open  **RS232 drive**, then find the sub-folder  **USB 2.0 TO RS232 Converter** and open it. Then chose  **driver**, and open it. This folder contains several drivers to be used in different operating systems. Simply install the suitable driver to the computer will do. (Note: during the installation, no serial cable should be connected, otherwise errors will occur.)
- 3) After installing the RS232 driver, connect a standard USB2.0 to RS232 serial cable between the computer and the laser. Then open the folder



and execute the software

C. Port Selection

Once the installation is complete, open the monitoring software which will show as in Figure 9. Choose the appropriate port on the left and then click the button on the right to enter the user interface.

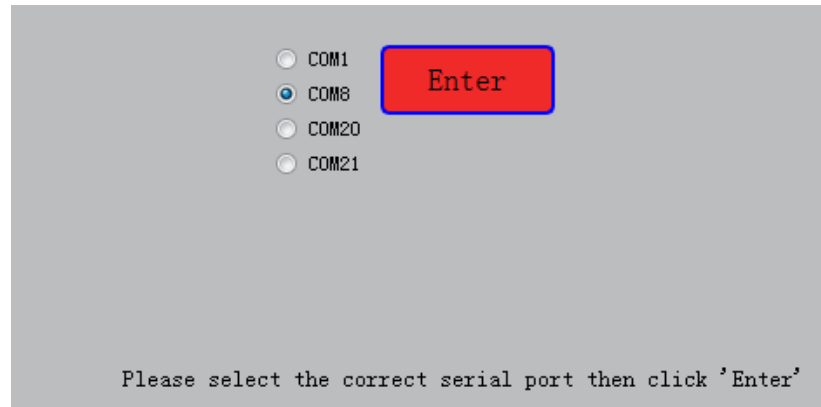


Figure 9 Serial port selection

D. Connection Status

Check whether the monitoring interface is connected properly, as shown in Figures 10 and 11.

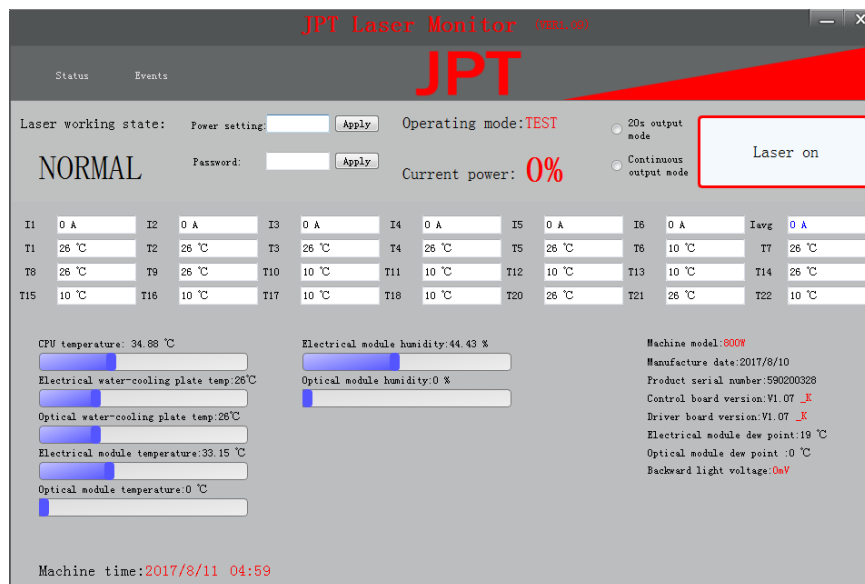


Figure 10 Properly connected

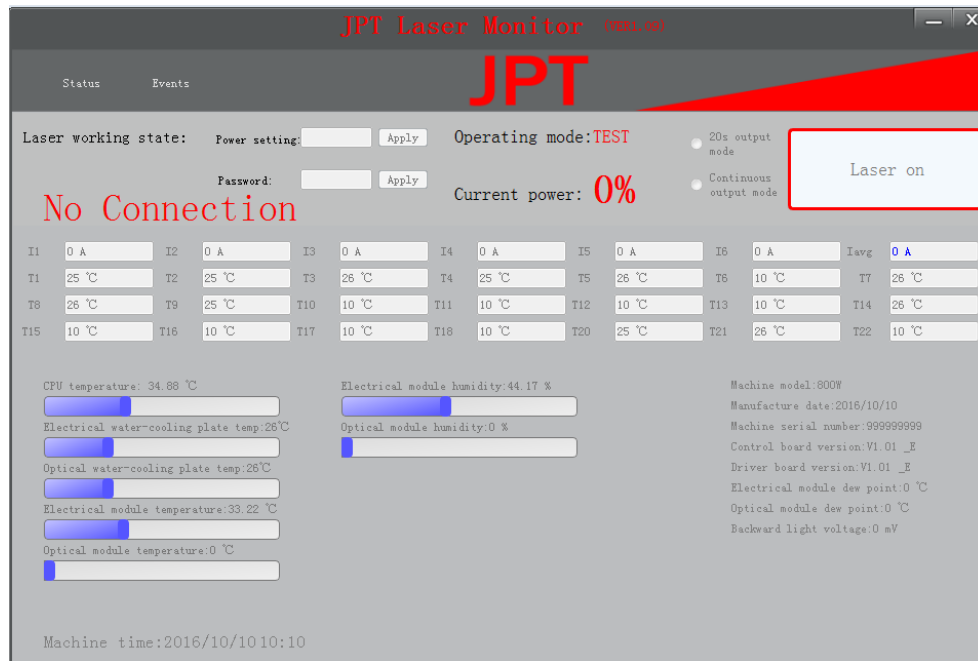


Figure 11 Disconnected

When you enter the interface, if it displays as shown in Figure 10, then it means the serial cable is connected properly and you can proceed to the next stage of operation.

When you enter the interface, if it displays as shown in Figure 11, then it means the serial cable is disconnected. Under this condition, the laser cannot be controlled through the software. There are 3 possible reasons for this: the laser is powered off, the standard USB2.0 to RS232 serial cable is damaged or wrongly connected. Please check for the reasons one by one and restart the operation after troubleshooting.

3) Laser State

A. Laser Mode Selection

This laser has two operating modes: Robot and Test.

✘ **Important: The laser cannot work until 5 seconds after it is powered on.**



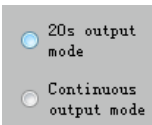
Figure 12 Mode selection

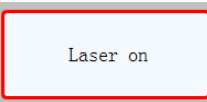
Test Mode: users can operate the laser using the controlling software.

Operating method: when the key switch is pointing at TEST on the front panel of the

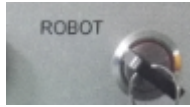
laser , Test Mode is selected and on the user interface, it will show

Operating mode: **TEST**. Set the power level using the software , and

then choose either 20S output or continuous-wave output . Once the setting is completed, press the "START" button at the front panel of the laser and the

laser will be ready to emit. Click  on the user interface will do.

Robot Mode: users can operate the laser from external control **Operating method:** when the key switch is pointing at ROBOT on the front panel of the laser



, Robot Mode is selected and on the user interface, , it will show

Operating mode: **ROBOT**

. Press the “START” button at the front panel of the laser and then once a signal is given from external the laser will emit. Emission under the Robot Mode needs to fulfill the following 4 conditions simultaneously:

- a) External 0-10V (Corresponding power ratio: 0-100%, Status can be monitored using the software; no emission when power is below 10%);
- b) Enable signal EN (External 24V);
- c) Modulation signal PWM (24V modulation signal source);
- d) START button or REDAY24V (START button switches the states and rising and falling edges of REDAY are also effective) ;

To achieve these four conditions, please refer to I/O definition and connect the cables accordingly.

B. Default Settings

Before leaving the factory, all the lasers have been set with the default parameters. There is no need and no way to change the settings. If under special requirement, please contact out technical staff.

5. Alarm Messages and Troubleshooting

1) Working Log Display

Laser emission under both of the modes can be monitored using the controlling software. It will send alarm messages when errors occur and will leave a record in the log, as shown in Figure 14.

The screenshot shows the JPT Laser Monitor software interface. At the top, the title bar reads "JPT Laser Monitor (32bit)". Below the title bar, there is a status bar with "Status" and "Events" tabs. The main area displays the JPT logo and various control elements. On the left, it says "Laser working state:" and "Power setting:" with an "Apply" button. In the center, it says "Operating mode: ROBOT" and "Current power: 79%". On the right, there are radio buttons for "20s output mode" and "Continuous output mode", and a "Laser on" button. A red "Alarm" message is displayed, stating "Sensor 1 alarm." Below this, there is a table titled "Event" with columns "Operation", "Time", and "Details".

Operation	Time	Details
Operation	08-09 15:09:23	Set the power to:79 %
Operation	08-09 15:09:26	Set the power to:80 %
Operation	08-09 15:09:29	Set the power to:79 %
Operation	08-09 15:09:32	Set the power to:80 %
Operation	08-09 15:09:40	Set the power to:79 %
Alarm	08-09 15:09:42	-Sensor 1 alarm--Please eliminate the alarm and restart the laser-
Operation	08-09 15:09:42	Set the power to:80 %
Operation	08-09 15:09:43	Laser off
Operation	08-09 15:09:45	Set the power to:79 %
Operation	08-09 15:09:48	Set the power to:80 %
Operation	08-09 15:09:54	Set the power to:79 %
Operation	08-09 15:09:56	Set the power to:80 %

Figure 14 Alarm logbook

2) Alarm Messages and Corrective Actions :

Table 10

S/N	Alarm Message	Phenomenon	Corrective Action	Note
1	Water cooling plate in the optical module is too hot	Laser emission is stopped, alarm buzzer is ringing, fault lamp is lit	1. Check whether the water supply to the laser is normal. 2. Check whether the water temperature setting of the chiller is within the regulated range.	After dismissing the alarm, restart the laser
2	Water cooling plate in the electrical module is too hot		As above	
3	QBH alarm		1. Check whether the connection between QBH and the cutting head is good.	
4	Water flow alarm		1. Check whether the inlet and outlet water pipes of the laser are connected properly. 2. Check whether the water supply channel is abnormal. 3. Check whether the water pressure in the chiller meet the requirement of the laser.	
5	Sensor 1 alarm		1. Observe whether there is red light when the alarm rings. If there is red light, then restart the laser and redo the operation. 2. If there is no red light, please contact the manufacturer. 3. If the alarm rings 3 times successively, the laser will be locked. Please contact the manufacturer directly.	
6	Emergency stop alarm		1. Check whether the emergency stop button is at the off state.	
7	Chiller fault alarm		1. Check whether the chiller is connected properly. 2. Check whether there is any problem with the chiller.	
8	The battery has been damaged, please contact the manufacturer		Please contact us directly.	

9	Sensor 1 failure		Please contact the manufacturer directly.	
10	Expired, please contact manufacturer		Please contact the manufacturer directly.	
11	Expired, please contact agency		Please contact the agency directly.	
12	Low pressure alarm		Please contact the manufacturer directly.	

Table 11

S/N	Alarm Message	Phenomenon	Corrective Action
1	Back reflection is too strong	Does not affect the laser emission.	When the alarm is dismissed, it can recover automatically. If this occurs when the user is cutting high-reflection material such as copper, the operation should be stopped immediately.
2	Dew formation on electrical module	Does not affect the laser emission.	When the alarm is dismissed, it can recover automatically.
3	Dew formation on electrical module	Does not affect the laser emission.	When the alarm is dismissed, it can recover automatically.
4	Sensor 2 failure	Does not affect the laser emission.	Restart the laser after the alarm is dismissed.
5	SD card failure	Does not affect the laser emission.	Restart the laser after the alarm is dismissed. Please contact us about the situation in time.

6. Maintenance, Customer Service and Repair

1) Warranty Information

When the user encounters problems with the laser which they cannot solve, please contact us and send the laser back for repair.

JPT will repair the products that prove to be defective during the warranty period. All products repaired or replaced under warranty are only warranted for the remaining period of time in the original warranty for the particular defective product.

2) Terms and Conditions

The above warranties do not apply to defects resulting from

- A. unauthorized modification, maintenance or repair;
- B. wrong operation, improper or inadequate maintenance or calibration by customer;
- C. improper use of operation outside of the specifications for the product;
- D. abuse, negligence, accident, loss or damage in transit;
- E. customer or third party supplied software, interfacing or supplies;

Upon receiving the product, customer has the obligation to read the user manual carefully and follow it strictly when operating the laser.

3) Customer Service and Repair

All the maintenance and repair should be done by the specialists in JPT. All requests for repair or replacement under this warranty must be made as soon as possible after the defect has been noticed and must be directed to JPT. If a refund is necessary and agreed between both parties, please verify that all components are on hand using the packing list and put the product and all the accessories back to the original package. If loss or damage occurs in transit because of the improper packaging done by the customer, it will not be covered by the warranty.

- ✘ **We reserve the right to make changes in design or construction of any of our products at any time without notice. All the specifications, parameters and agreement are subject to the final sales contract.**
- ✘ **The interpretation of the Terms and Conditions belongs to Shenzhen JPT Opto-electronics Co., Ltd.**